

Truong Hoai An Luu

12/11/2025

EE303

Final Exam

1. I have submitted the assessment and survey to earn credit.

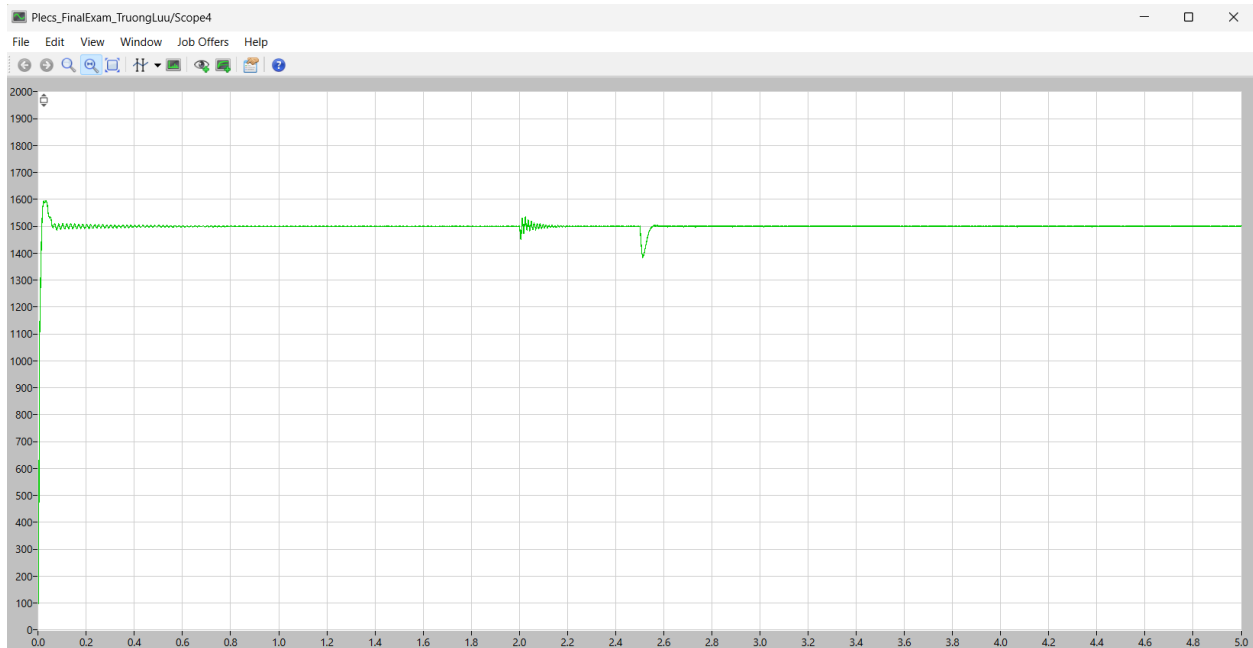
2. I confirm that I have not discussed nor collaborated with other persons related or unrelated to EE 3030 to generate the answers to this final exam. I understand that if cheating is detected, I will receive a zero grade in my final exam.

Signature: TRUONG HOAI AN LUU

3. Implement:

a. The boost converter:

The DC voltage at the DC side of the converter v_{dc}



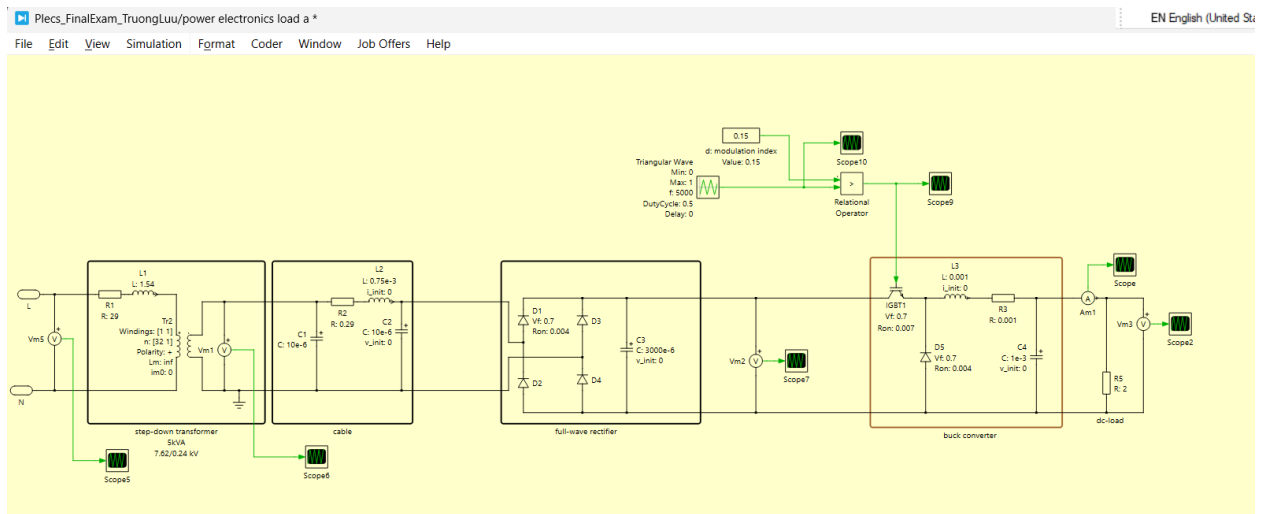
(Fluctuations of less than 100 V around the nominal 1500 Vdc)

b. The three power electronics loads (a, b, and c):

The voltage trace at the low DC voltage side of the buck converter



(Sinusoidal fluctuations of less than 5 volts are expected around 48 Vdc)



(This photo shows my modulation index value)

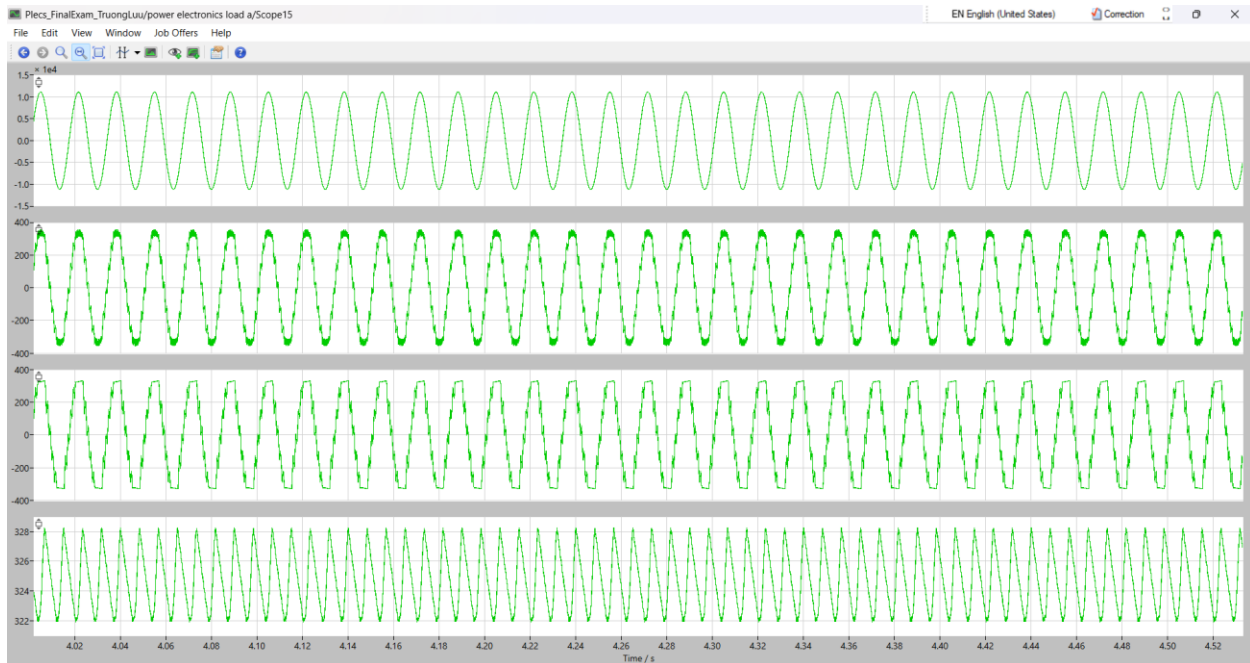
4. Traces of the voltages, currents, and average power:

Traces of the voltages:

+ Voltage at Point 1 looks like the AC input waveform. Point 2 is still AC, but its magnitude reflects the transformer stage.

+At Point 3, the waveform no longer behaves like a clean bipolar sine, but its shape now indicates the signal has been rectified, producing a mostly one-polarity waveform with ripple.

+Finally, Point 4 is the DC of the four: it is largely flat compared to Points 1–3, but you can still see high-frequency ripple, which is consistent with the buck converter output.



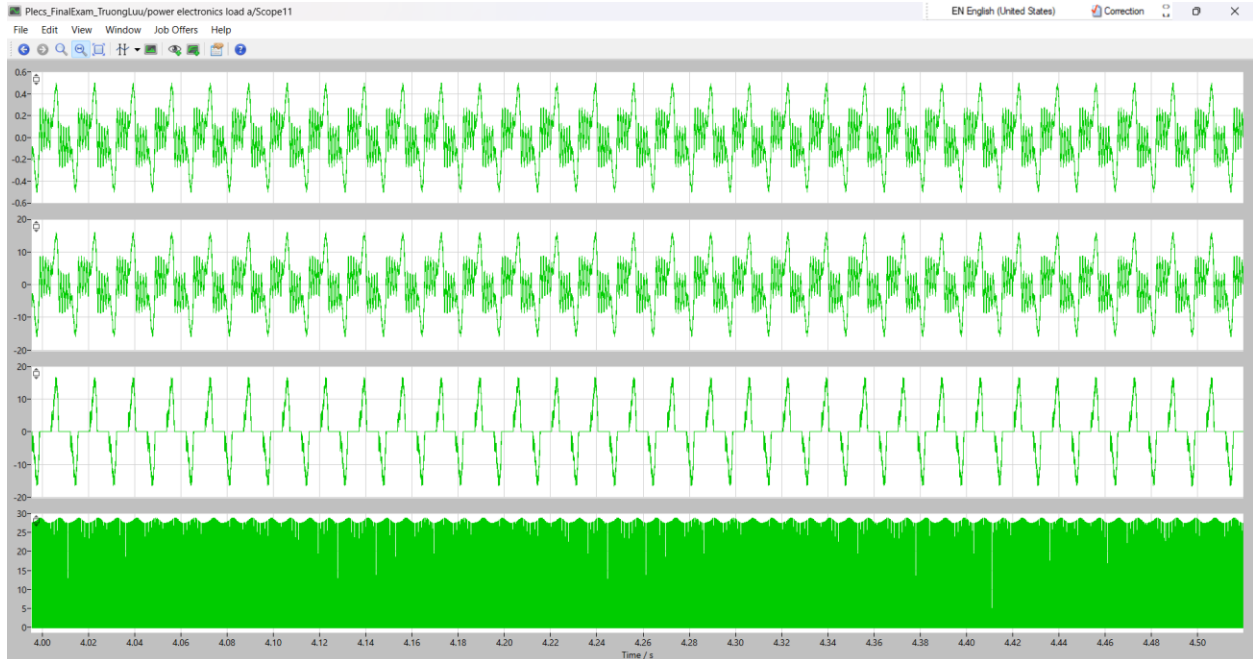
(Traces of voltages for $t \in [4.0, 4.5]$ s)

Traces of the currents:

+ Currents at Points 1 and 2 are very similar overall, which makes sense because they are tied to the same upstream AC behavior.

+ Point 3 current is more discontinuous, matching what to expect after rectification, where conduction happens in bursts rather than smooth sinusoidal.

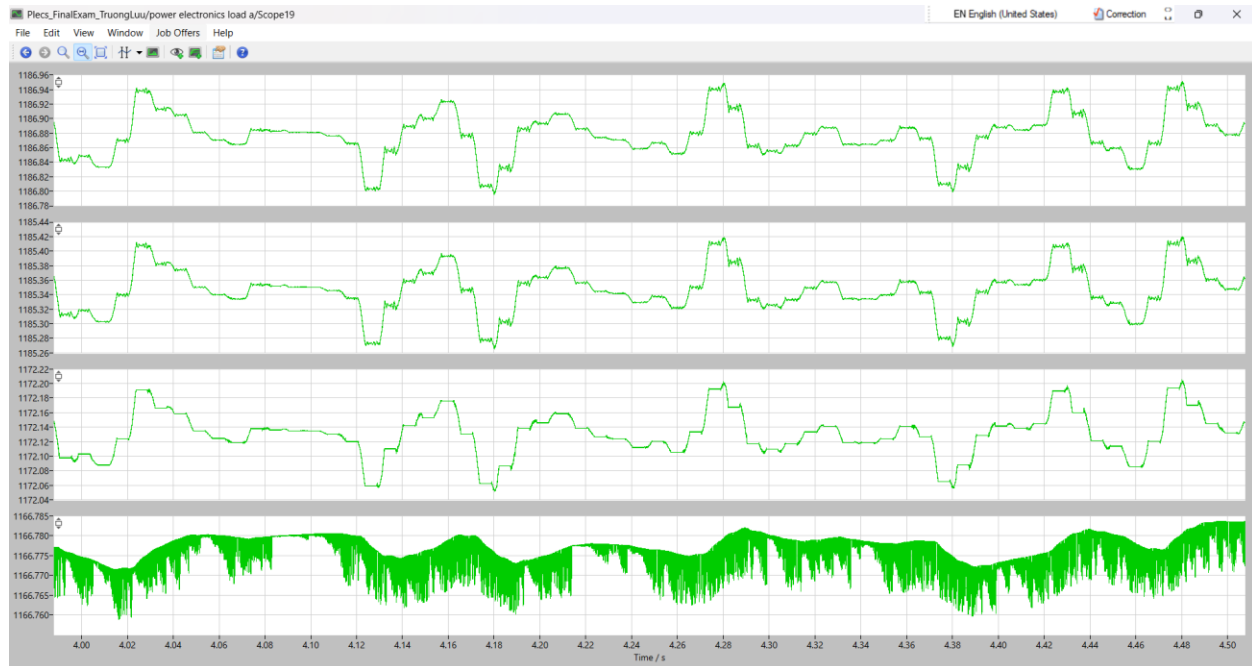
+ Point 4 current: it appears much denser and more ripple-heavy, consistent with a converter stage, where the waveform includes significant high-frequency components even though the output is delivering steady power.



(Traces of current for $t \in [4.0, 4.5]$ s)

Traces of the average power:

- + The average power at Points 1–3 follows a very similar trend and sits in the same general range, meaning most of the input power is being transferred through the transformer stages with comparable average levels.
- + Point 4 behaves differently because it settles to a lower average level than Points 1–3 and shows a more noticeable ripple in the averaged trace, which matches the idea that the buck converter regulates the downstream load and changes the voltage/current levels compared to the upstream points.

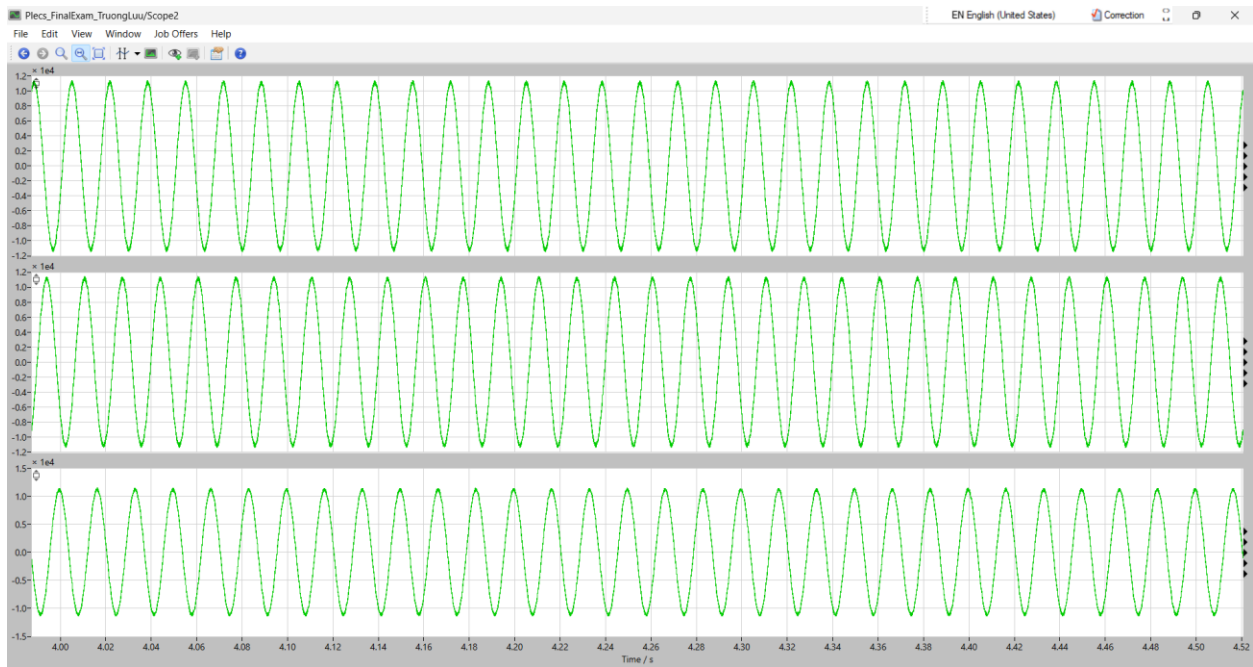
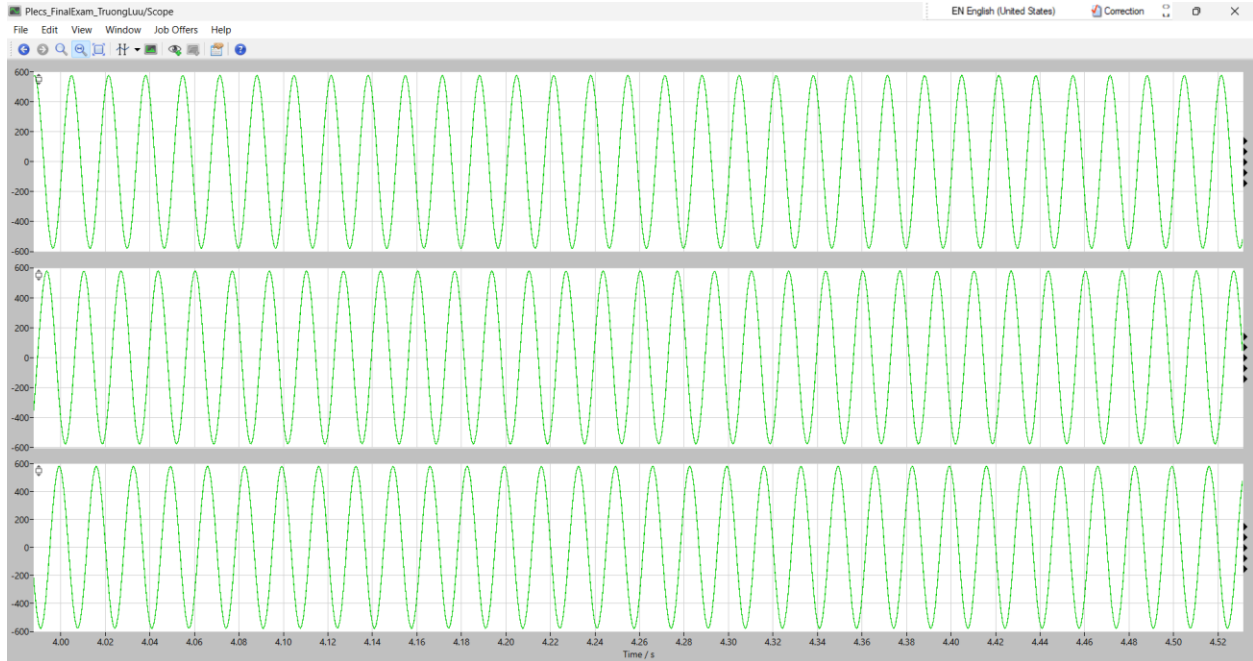


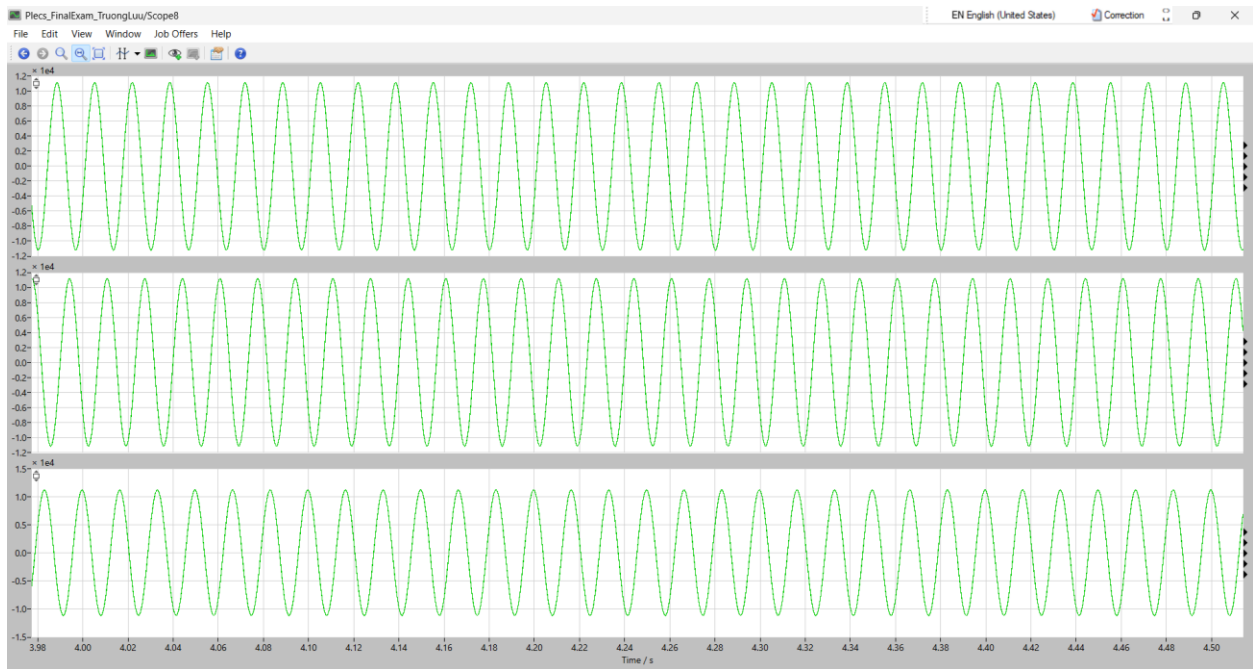
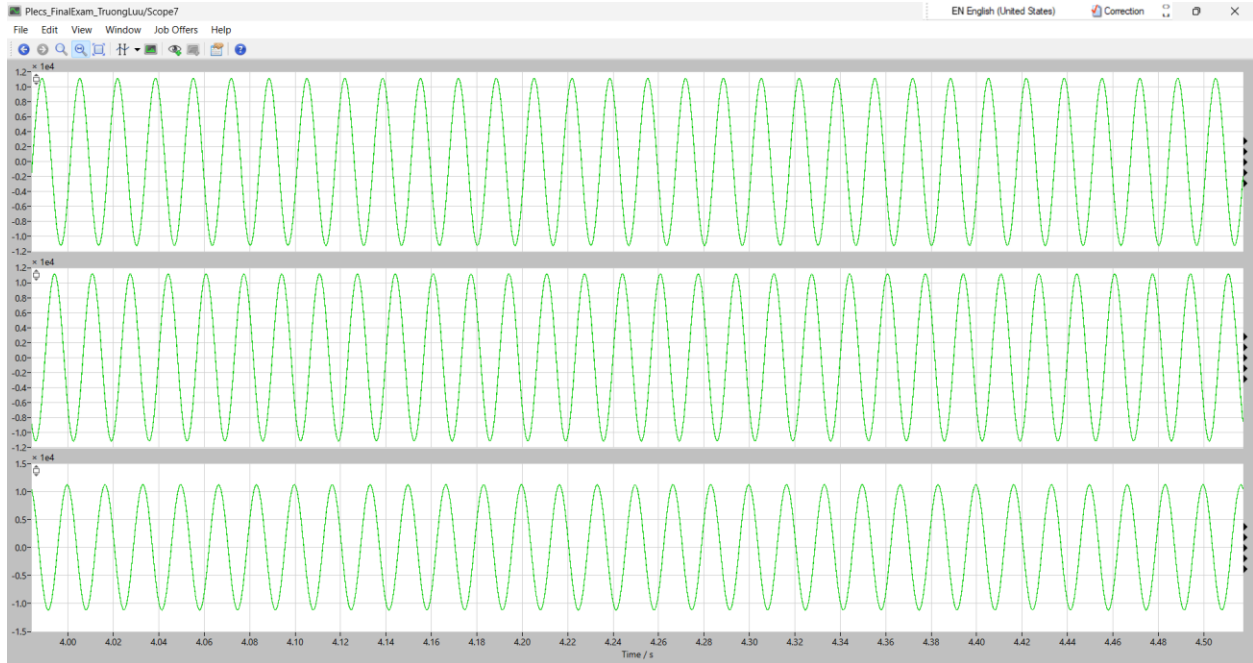
(Traces of average power for $t \in [4.0, 4.5]$ s)

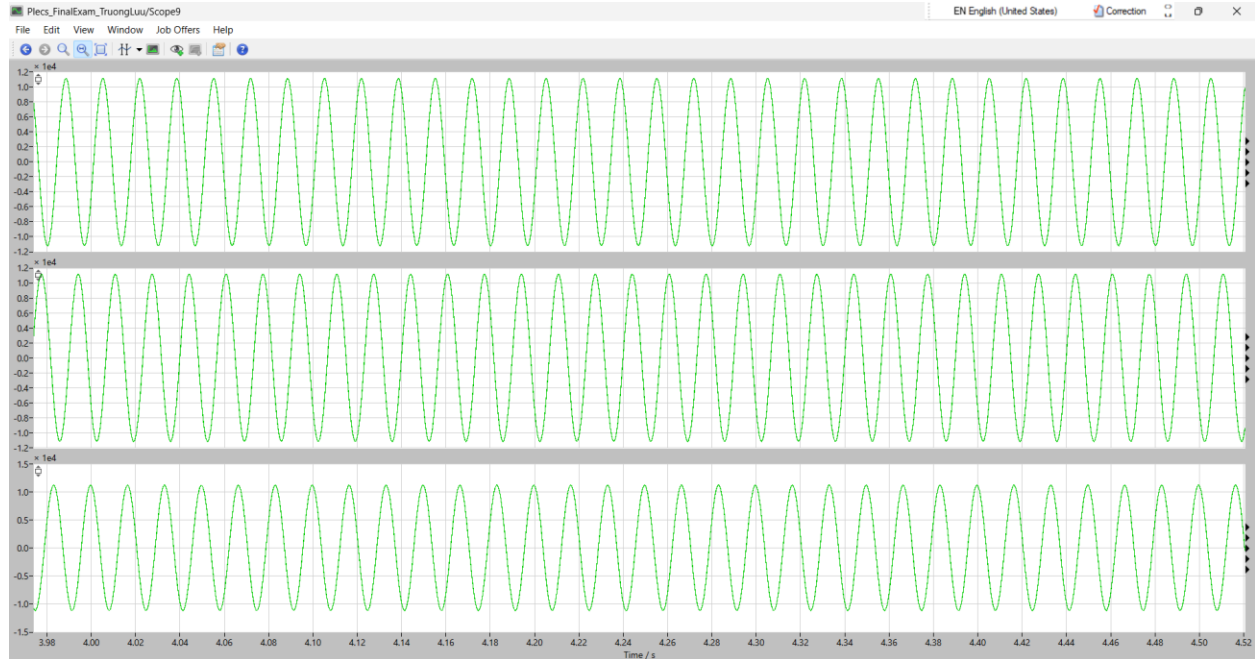
5. Waveforms of the line-to-ground voltages, three-phase currents, and average power ($t \in [4.0, 4.5]$ s):

The line-to-ground voltages:

+ 1-5 all look the same.



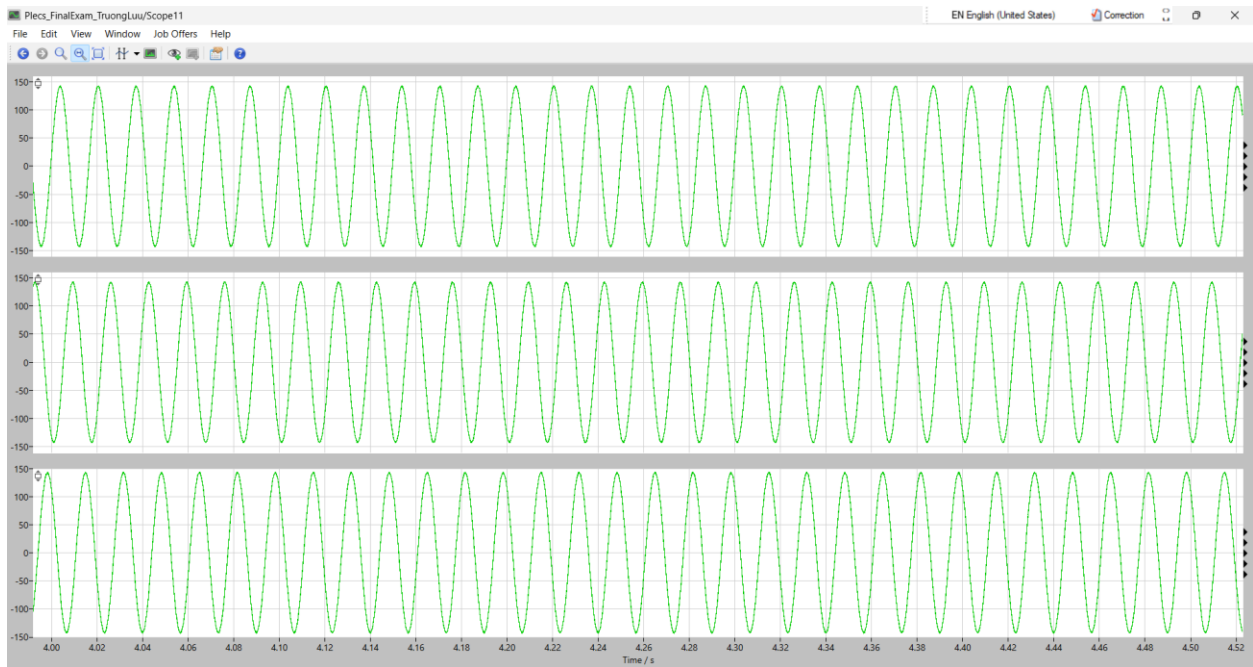
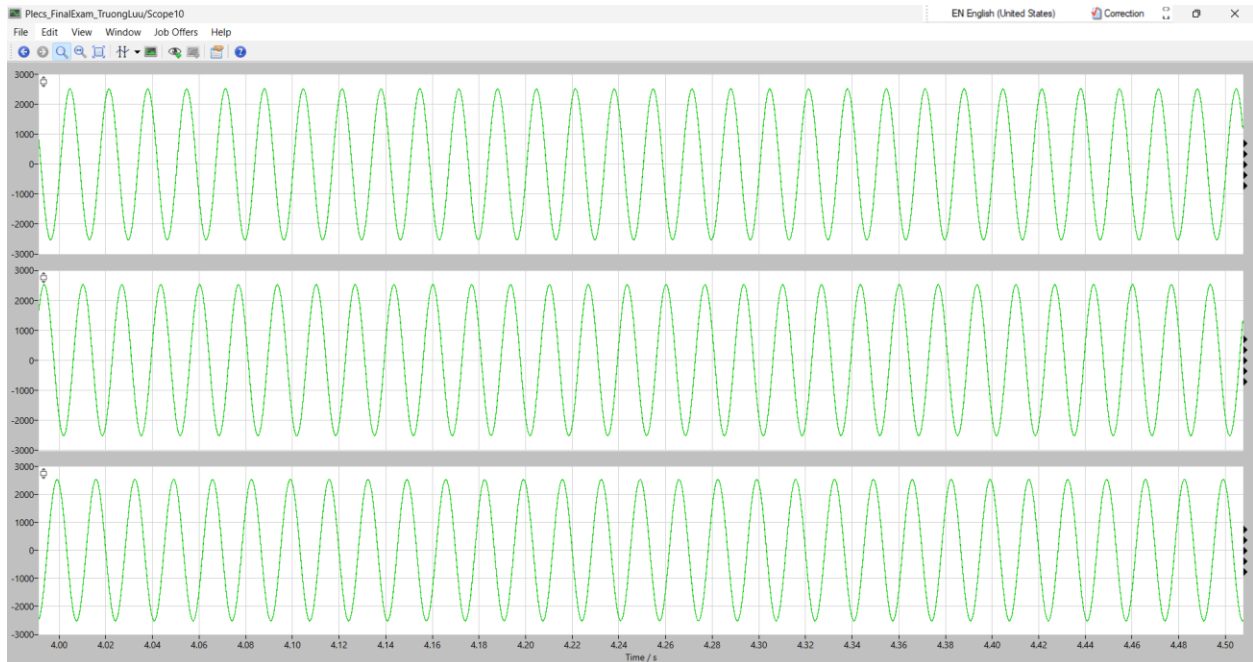


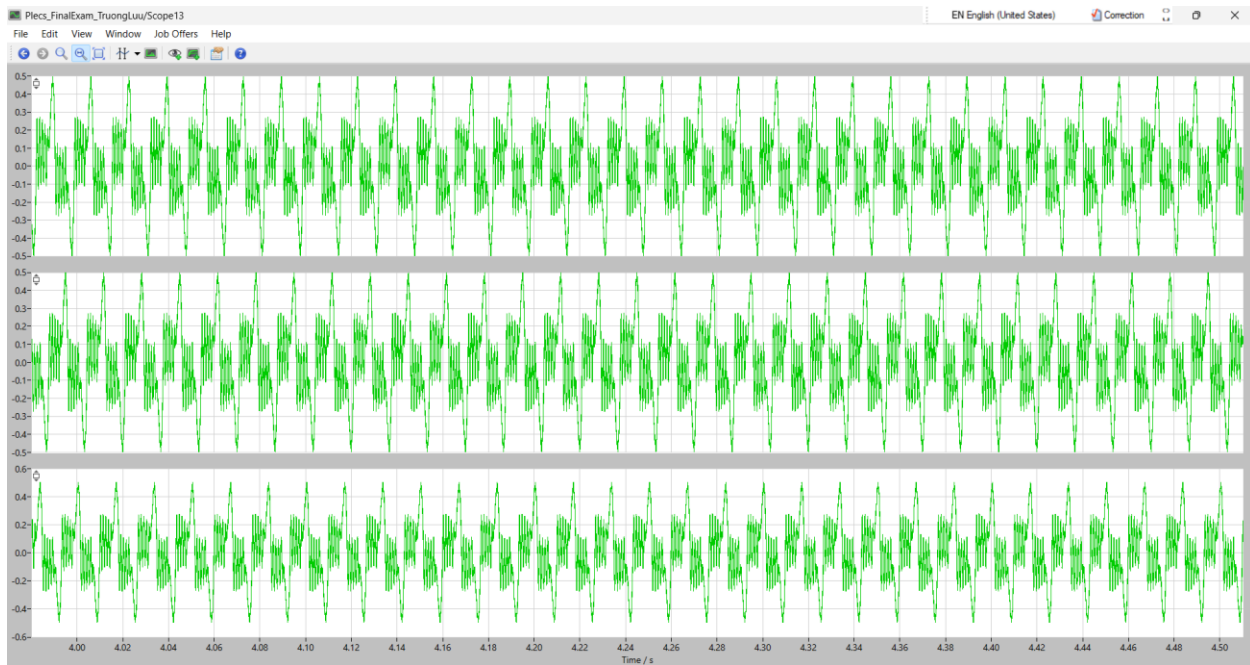
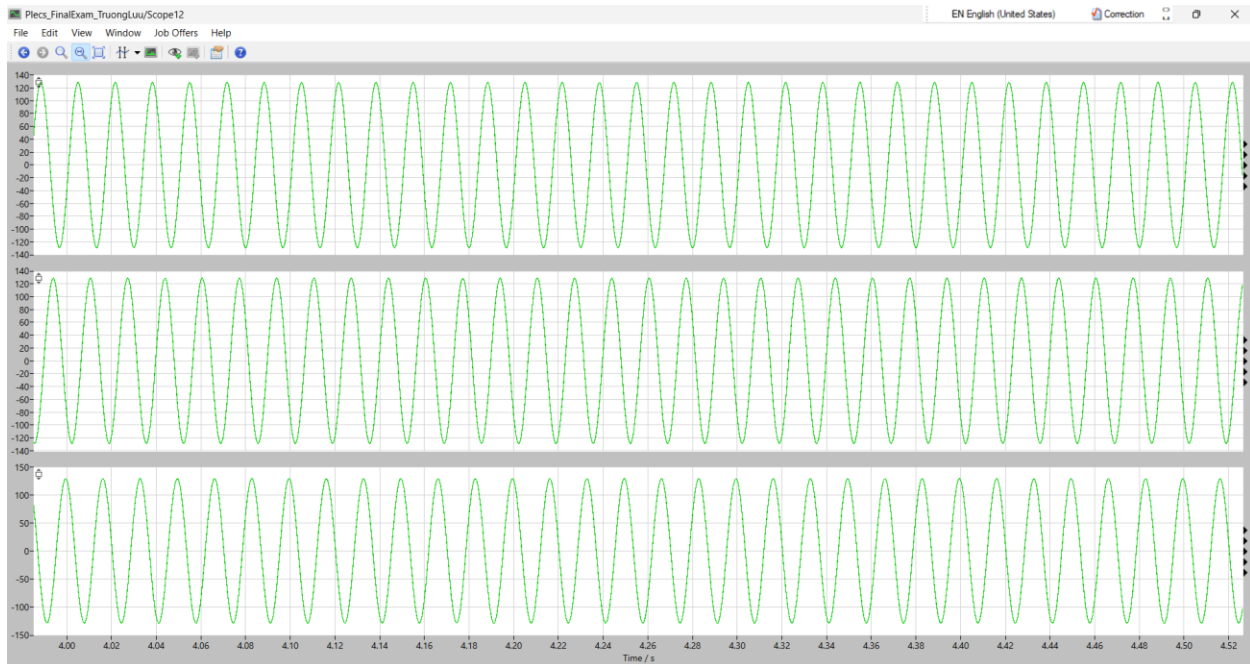


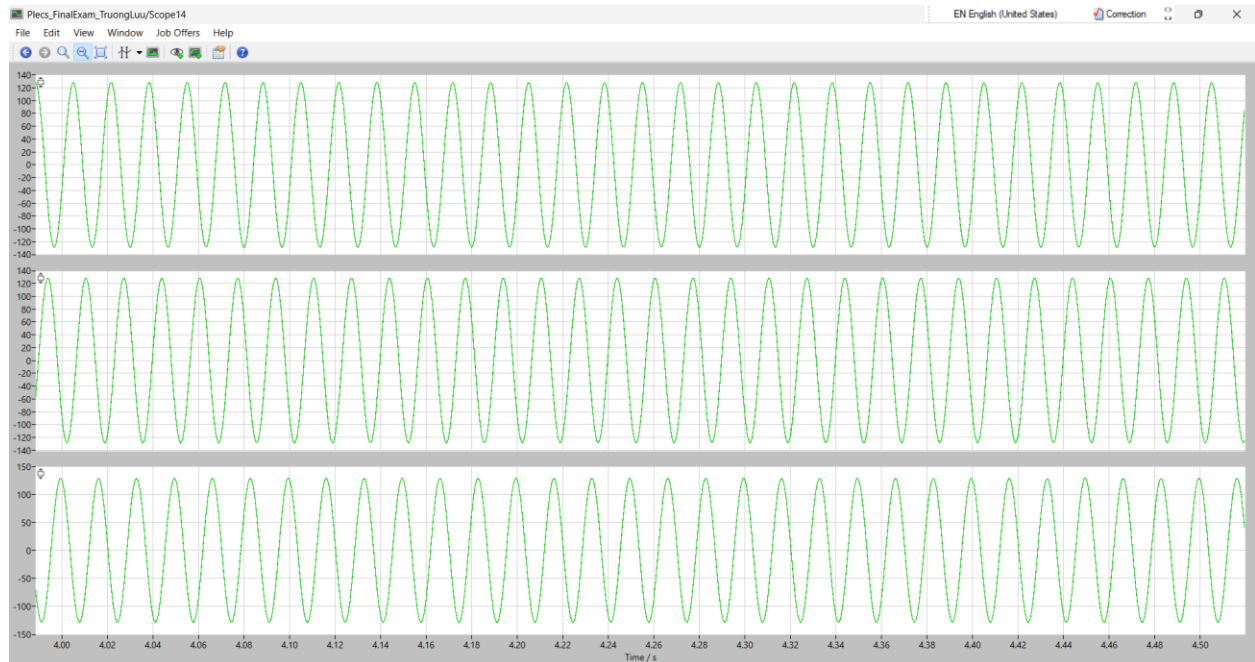
Three-phase currents:

+ 1-3, and 5 look the same.

+ 4 is completely different compared to the rest.







Average power:

+ 1, and 3-5 look the same.

+ The waveform of 2 looks thicker than the rest.

